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ROLL NO:- 2520030188

SEC NO:- 02

1.Practical5b.c

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int fd[2];

    pipe(fd);

    if(fork()==0)
    {
        dup2(fd[1], STDOUT_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("ls", "ls", "-l", NULL);
    }

    if(fork()==0)
    {
        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("grep", "grep", ".c", NULL);
    }

    close(fd[0]);
    close(fd[1]);

    wait(NULL);
    wait(NULL);

    return 0;
}
```

OUTPUT:-



praneeth@PRANEETH: ~



```
praneeth@PRANEETH:~$ nano pipe1.c
```

```
praneeth@PRANEETH:~$ gcc pipe1.c -o pipe1
```

```
praneeth@PRANEETH:~$ ./pipe1
```

```
total 676
```

```
-rwxr-xr-x 1 praneeth praneeth 16304 Aug  8 03:59 1
-rw-r--r-- 1 praneeth praneeth  495 Aug  8 03:56 1.c
-rwxr-xr-x 1 praneeth praneeth 16168 Aug  8 03:34 2
-rw-r--r-- 1 praneeth praneeth  457 Aug  6 04:06 2.c
-rw-r--r-- 1 praneeth praneeth   82 Jul 30 08:36 P3.c
-rwxr-xr-x 1 praneeth praneeth 16384 Sep  2 05:25 SIGCHILD
-rw-r--r-- 1 praneeth praneeth  536 Sep  2 05:25 SIGCHILD.c
-rwxr-xr-x 1 praneeth praneeth 16096 Aug 28 06:47 fcfc
-rwxr-xr-x 1 praneeth praneeth 16096 Aug 28 06:47 fcfs
-rw-r--r-- 1 praneeth praneeth  583 Aug 28 06:47 fcfs.c
-rwxr-xr-x 1 praneeth praneeth 16208 Aug 21 15:30 fexe
-rw-r--r-- 1 praneeth praneeth  513 Aug 21 15:28 fexe.c
-rwxr-xr-x 1 praneeth praneeth 16040 Aug 27 04:00 fifo
-rw-r--r-- 1 praneeth praneeth  197 Aug 27 04:00 fifo.c
-rwxr-xr-x 1 praneeth praneeth 16040 Aug 27 04:06 fifo_read
-rw-r--r-- 1 praneeth praneeth  197 Aug 27 04:06 fifo_read.c
-rwxr-xr-x 1 praneeth praneeth 16176 Aug 27 06:51 fifo_write
-rw-r--r-- 1 praneeth praneeth  272 Aug 27 04:19 fifo_write.c
-rwxr-xr-x 1 praneeth praneeth 16312 Aug 21 15:34 keyb
-rw-r--r-- 1 praneeth praneeth  707 Aug 21 15:33 keyb.c
-rw----- 1 praneeth praneeth  707 Aug 21 15:38 keyb1.save
prw-r--r-- 1 praneeth praneeth    0 Aug 27 06:53 myfifo
drwxr-xr-x 2 praneeth praneeth  4096 Jul 30 04:18 os
-rw-r--r-- 1 praneeth praneeth  100 Jul 30 09:04 p3.c
-rwxr-xr-x 1 praneeth praneeth 15992 Jul 30 09:11 p4
-rw-r--r-- 1 praneeth praneeth  100 Jul 30 09:11 p4.c
-rwxr-xr-x 1 praneeth praneeth 15992 Jul 30 09:38 p5
-rw-r--r-- 1 praneeth praneeth   90 Jul 30 09:14 p5.c
-rwxr-xr-x 1 praneeth praneeth 15992 Aug  1 05:47 p6
-rw-r--r-- 1 praneeth praneeth  175 Aug  1 05:45 p6.c
-rwxr-xr-x 1 praneeth praneeth 15992 Jul 30 09:19 p6a
-rw-r--r-- 1 praneeth praneeth   97 Jul 30 09:19 p6a.c
-rwxr-xr-x 1 praneeth praneeth 16032 Aug  1 05:56 p7
-rw-r--r-- 1 praneeth praneeth  174 Aug  1 05:56 p7.c
-rwxr-xr-x 1 praneeth praneeth 16032 Aug  1 09:34 p8
-rw-r--r-- 1 praneeth praneeth  210 Aug  1 09:34 p8.c
-rwxr-xr-x 1 praneeth praneeth 16208 Sep  3 05:59 pipe1
-rw-r--r-- 1 praneeth praneeth  522 Sep  3 05:58 pipe1.c
-rwxr-xr-x 1 praneeth praneeth 16296 Aug 27 03:38 pipes
-rw-r--r-- 1 praneeth praneeth  469 Aug 27 03:32 pipes.c
```

```

praneeth@PRANEETH: ~
-rw-r--r-- 1 praneeth praneeth 1251 Aug 28 06:57 round.c
-rwxr-xr-x 1 praneeth praneeth 16184 Sep  3 04:00 serverclient
-rw-r--r-- 1 praneeth praneeth  645 Sep  3 03:59 serverclient.c
-rwxr-xr-x 1 praneeth praneeth 16216 Aug 21 14:42 shell
-rw-r--r-- 1 praneeth praneeth 16216 Aug 21 14:44 shell.c
-rwxr-xr-x 1 praneeth praneeth 16304 Aug 21 15:32 shll
-rw-r--r-- 1 praneeth praneeth  463 Aug 21 15:31 shll.C
-rw-r--r-- 1 praneeth praneeth  463 Aug 21 15:32 shll.c
-rwxr-xr-x 1 praneeth praneeth 16112 Aug 29 05:18 signal
-rw-r--r-- 1 praneeth praneeth  281 Aug 29 05:18 signal.c
-rwxr-xr-x 1 praneeth praneeth 16168 Sep  2 05:58 sigterm
-rw-r--r-- 1 praneeth praneeth  342 Sep  2 05:13 sigterm.c
-rwxr-xr-x 1 praneeth praneeth 16192 Aug 22 09:25 simplecommandloop
-rw-r--r-- 1 praneeth praneeth  303 Aug 22 05:34 simplecommandloop.c
-rwxr-xr-x 1 praneeth praneeth 16096 Aug 28 06:50 sjf
-rw-r--r-- 1 praneeth praneeth  845 Aug 28 06:50 sjf.c
-rwxr-xr-x 1 praneeth praneeth 16216 Aug 26 09:32 skill2
-rw-r--r-- 1 praneeth praneeth  460 Aug 26 09:31 skill2.c
-rwxr-xr-x 1 praneeth praneeth 16344 Aug 26 09:14 skill3
-rw-r--r-- 1 praneeth praneeth  806 Aug 26 09:04 skill3.C
-rw-r--r-- 1 praneeth praneeth  806 Aug 26 09:04 skill3.c
-rw----- 1 praneeth praneeth    0 Aug 26 06:08 skill3.save
-rwxr-xr-x 1 praneeth praneeth 16216 Aug 26 09:21 skill4
-rw-r--r-- 1 praneeth praneeth  460 Aug 26 09:20 skill4.c
-rwxr-xr-x 1 praneeth praneeth 16352 Sep  3 03:59 socketserver
-rw-r--r-- 1 praneeth praneeth  922 Sep  3 03:57 socketserver.c
-rwxr-xr-x 1 praneeth praneeth 16040 Aug 22 04:52 zombie
-rw----- 1 praneeth praneeth   67 Aug 22 04:42 zombie.save
-rw-r--r-- 1 praneeth praneeth  189 Aug 22 04:51 zomibe.c
praneeth@PRANEETH:~$

```

2. Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers.

```
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handler(int sig)
{
    if(sig==SIGINT)
        printf("\nSIGINT Received\n");

    if(sig==SIGTERM)
        printf("\nSIGTERM Received\n");

    if(sig==SIGUSR1)
        printf("\nSIGUSR1 Received\n");
}

int main()
{
    signal(SIGINT,handler);
    signal(SIGTERM,handler);
    signal(SIGUSR1,handler);

    while(1)
    {
        printf("Running...\n");
        sleep(3);
    }

    return 0;
}
```

OUTPUT:-

(TERMINAL -1):-

```
praneeth@PRANEETH: ~  
praneeth@PRANEETH:~$ nano pipe2.c  
praneeth@PRANEETH:~$ gcc pipe2.c -o pipe2  
praneeth@PRANEETH:~$ ./pipe2  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...  
^C  
SIGINT Received  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...
```

(TERMINAL :-2)

```
C:\Users\praneeth>wsl  
praneeth@PRANEETH:/mnt/c/Users/praneeth$ pgrep pipe2  
632  
praneeth@PRANEETH:/mnt/c/Users/praneeth$ kill -SIGUSR1 632  
praneeth@PRANEETH:/mnt/c/Users/praneeth$ |
```

AGAIN TERMINAL 1:-

×

+

[illegible]

SIGUSR1 Received

[illegible] $\wedge C$

SIGINT Received

Running...

[^]Z

[1]+ Stopped

```
./pipe2
```

```
praneeth@PRANEETH:~$
```